Two-tail t critical/p-value method + t test ↔ CI

A1S2: SLIDES 1-51 ONLY

1 RECOGNISE IT

- Two-tail mean test; σ unknown; t assumptions acceptable.
- Need critical values, p-value and/or matching CI conclusion.
- This is the final in-scope Chapter 9 method family.

2 DECISION / METHOD

1. tcalc=($\bar{X}$-μ_0)/(s/√n); df=n-1.
2. Critical route: compare |tcalc| with T.INV.2T(α,df).
3. p route: T.DIST.2T(ABS(tcalc),df).
4. Reject iff p<α.
5. Matching t CI excludes μ_0 iff test rejects.

3 FORMULA

```
t* = T.INV.2T(α,df) > 0
two-tail p=T.DIST.2T(|tcalc|,df)
CI: X¯ ± t* × s/√n
```

4 FORMULA SHEET VS WHAT I KNOW

Gives: t lane, df and SE.
Still know: t critical/p-value functions and two-tail CI equivalence.

5 EXCEL ROUTE

- `tstar=T.INV.2T(alpha,df)`
- `p=T.DIST.2T(ABS(tcalc),df)`
- CI endpoints=`Xbar +/- tstar*s/SQRT(n)`.

6 TINY WORKED EXAMPLE

$\bar{X}$=18.4,s=2.5,n=25,μ_0=17: tcalc=2.8,df=24,p about .0099. At α=.05 reject; matching CI (17.368,19.432) excludes 17.

7 TOP EXAM TRAPS

- Using T.DIST instead of T.DIST.2T for a two-tail p-value.
- Using NORM.S.INV for t critical.
- Comparing μ_0 with a prediction range instead of mean CI.

8 SAY ALOUD

For a two-tail t test, T.DIST.2T gives the p-value and the matching t interval gives the same decision.

9 EXIT CHECK

Which Excel function returns the two-tail t p-value?

ONE-GLANCE VISUAL ROUTE

